Towards Adaptive Multimedia System for Assisting Children with Arabic Learning Difficulties

Moutaz Saleh, Jihad Mohamad Alja'am
Department of Computer Science & Engineering
College of Engineering, Qatar University
Doha, Qatar
{moutaz.saleh, jaam}@qu.edu.qa

***Abstract*—Children with learning difficulties (LD) are increasing dramatically in the Arab world. Such children require quick intervention especially during the early childhood years. This paper presents a dynamic multimedia system for helping children with LD to overcome their learning problems. We introduce an approach to automatically convert modern standard Arabic children's stories to the finest representative images that can efficiently illustrate the meaning of words. Specifically, first, we apply natural language processing techniques to analyze the text in stories and we extract keywords of all characters and events in each sentence. Second, we apply an image captioning process through a pre-trained deep learning model for all retrieved images from our multimedia database as well as the Google search engine. Third, using sentence similarities, most significant images are retrieved back by selecting top highest similarity values. The proposed system aims to better enhance understanding, communications, and thinking skills for children with LD in elementary schools and special education centers.**

Keywords—Learning; Assistive Technology; Multimedia Content; Knowledge Extraction, Semantic Analysis, Natural Language Processing.

I. INTRODUCTION

Generally, children with LD have problems in several language skills [1,2], for instance they cannot distinguish properly among letters and words; different words; forming simple proper sentences. In addition, they may have difficulties in understanding the correct meaning of simple words in each context, forgetting names of objects, establishing logical relations between words. These difficulties can be due to different physical and psychological problems, i.e. intellectual disabilities, concentration deficit, difficulties in learning through conventional teaching methods, disorganization, limited self-reliance, and low self-esteem.

Children with LD are of average intelligence and they need different teaching methodologies to help them understand. They can continue as normal students if the appropriate teaching methods are found to suit their effective needs. These children are currently integrated in normal schools, and this adds key strains on these schools and its teachers. In fact, these children need more dedication and one-to-one teaching methodology, but schools cannot recruit new teachers to follow children individually. Indeed, Early intervention with appropriate teaching and assessment methods tailored to these children's effective needs can decrease their delays in learning and break their isolation. The interactivity between teacher and student is important for learning but is almost absent in the learning sessions due to the children's lack of understanding of the concepts being taught, and to the teachers' lack of media resources in remedial teaching. Multimedia technology can improve language learning for children with LD, it keeps the children engaged for a longer time and takes into consideration their different levels of difficulties to learn new concepts.

Recently, Increasing attention has been drawn in the Human-Computer Interaction (HCI) community towards the design and development of accessible computer applications for children with LD. Due to better healthcare and education, the learning quality of for such children has been obviously improved especially that these children grow up and live in the community while surrounded by computer applications and software. Despite the potential benefit of the usage of assistive computer technology for children with LD and the potential that this technology may provide for them, there is a substantial need of baseline information on a number of topics, including the basic computer skills of people with LD, the objectives for using computers, and the difficulties they experience while using computers. Knowledge on these issues will help researchers and practitioners to form an accurate picture of this unique population, and make better use of computer technology.

In this paper, we present an adaptive multimedia learning system for assisting children with Arabic LD to enhance their understanding, communications, and thinking skills. Our proposed system dynamically generates multimedia contents which can be effectively used by instructors to provide children with LD the opportunity to learn with contents designed to suit their intellectual needs, and maximize the education process through reducing the administrative and teaching overheads. This paper is organized as follows: section 2 introduces related works, section 3 summarizes text-to-picture systems that have been developed, section 4 describes our adaptive multimedia contents development showing both Arabic text processing and its knowledge extraction, section 5 shows system operation, section 6 presents the assessment method, while section 7 concludes the paper.

II. RELATED WORK

Researchers in special education are to some extent informed of the potential of computer technology in helping children with LD [3]. The claim is that computer technology can help such children increase confidence and motivation through creative activities. Computer technology also has other benefits, including errorless learning, immediate feedback, self-paced learning, and independence of learning. Though, it is widely advised that the actual benefits of computer technology may be reduced or not even apparent depending on the quality of the

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software, examples are the contents of many software programs are not age appropriate, many educational software are unable to reach educational goals and are used as a tool for mere entertainment, and many applications do not promote independent learning. Thus, It is important to examine how assistive technologies could help children with LD be more independent, yet remain connected to others and in touch to get assistance when needed.

There have been many efforts in developing multimedia learning systems in the literature. Cheng et al. [3] designed an online learning system for Arabic language with ready-made content. The system can select the learning material for every learner based on the knowledge that he provides at start. Erradi et al. proposed a simple game-like system called "ArabicTutor" [4] to teach Arabic. The content shows an Arabic word with its synonyms in different contexts with images. Wastam et al. [5] have developed a system to teach children stories through flashcards. The instructor selects a story and asks then the child to arrange its flashcards in a logical sequence. Rosmani and Abdul Wahab [6] proposed a simple prototype called "i-IQRA" to teach children the Holy Qura'n. It helps them to pronounce the verses in the right manner. Cheng et al. [7] proposed Crome for children and adults. Every lesson is followed by a set of exercises to assess the progress of the learners. Tabot and Hamada [8] have proposed a web-based educational system to teach physics to students. Ping et al. [9] have built an educational system for children with hearing impairment to teach them the Malay language. Wuang et al. [10] have built a multimedia courseware system that is based on learning theories. All these systems are based on static contents and address different learning objectives. However, none of the related work that we have seen so far could adaptively converting Arabic language text into an appropriate personalized multimedia contents that would significantly enhance the learning process for children with LD.

III. TEXT-TO-PICTURE SYSTEMS

A Text-to-picture system is a system that automatically converts natural language text into pictures representing the meaning of that text. The pictures can be static illustrations such as images or dynamic illustrations such as animations. In this section, we investigated and analyzed a number of well-known text-to-picture systems and approaches. We show their constituent steps such as knowledge extraction, mapping, image layout, as well as their performances and limits.

A. Story Picturing Engine

This system refers to the process of illustrating a story with suitable pictures [11]. The system is a pipeline of three processes namely: story processing and image selection, estimation of similarity, and reinforcement-based rank. The estimation of similarity between pairs of images is based on their visual and lexical features. The images are ranked based on a mutual reinforcement method and most highly-ranked images are retrieved. Figure 1 shows an output for a story picturing engine.



Figure 1. Output from Story Picturing Engine for the text "on walk through Paris", H = highest ranked, L = Lowest ranked

Despite the good accuracy and performance of the story picturing engine, it only retrieves one picture for a given story and ignores many aspects such as the temporal or spatial relations. More advanced language processing techniques can be incorporated into the Story Picturing Engine for richer performance.

B. Text-to-Picture Synthesis System

In contrast to the story picturing engine, this system [12] associates a different picture to each extracted key phrase and presents the story as a sequence of related pictures (see Figure 3). It treats the text-to-picture conversion problem as an optimization process but still inherits the drawbacks of text-to-picture systems despite its performance compared to Story Picturing Engine.

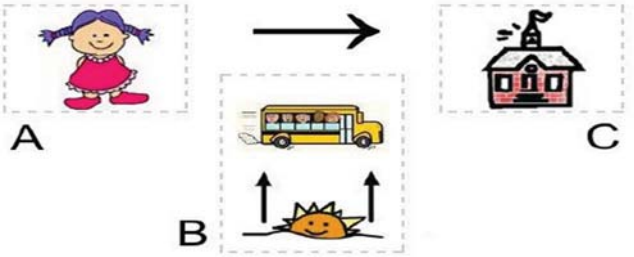


Figure 2. Output from Text-to-Picture Synthesis for the sentence "The girl rides the bus to school in the morning"

C. Word2Image

This system proposed various techniques [13], including the correlation analysis, semantic and visual clustering in order to produce sets of images from Flickr. To make the representative image more easily understandable, a large version of the original sized image will be shown when the user places the mouse over it, and the top-4 keywords will be displayed to depict its content. Figure 3 shows an example output for the concept "pyramid".



Figure 3. Output from Word2Image for the word "Pyramid"

Although, Word2Image translate concepts into its visual counterpart with sets of high quality, precise, diverse and representative images, it inherits the problems related to input language such as restricting the input to single words only.

D. Chat with Illustration

This is a visual assisted instant messaging scheme named Chat with illustration (CWI), which presents users visual messages associated with textual message automatically [14]. The CWI scheme consists of Image database construction, Dialogue analysis, Image selection and ranking, Visual layout, Visual dialogue summarization. Figure 4 shows visualized dialogue. Despite the accuracy and the intuitiveness of the visual layout, CWI relies on making excessive preparation of image resource. Furthermore, only few unpicturable concepts, such as verbs, adjectives, fixed phrases and interrogatives are considered and labeled to images manually which makes the chat directed more to concrete concepts only. Another disadvantage of this system

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is its limited language processing capabilities and grammatical and logical relations.



Figure 4. Output for the message "Good idea."

E. *WordsEye*

WordsEye is a text-to-scene system that can convert input text into representative 3D static scenes automatically [15]. The system consists of two main components, including a linguistic analyzer and a scene depicter. The geometric information of the objects is attached to 3D models, and once the layout is completed, the static scene is rendered using OpenGL similar to the example shown in Figure 5.

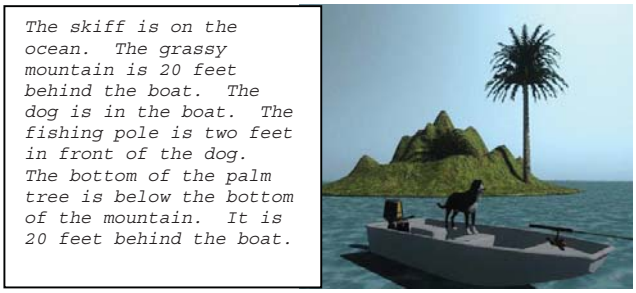


Figure 5. Output for the text on the left side

Although WordsEye has achieved a good degree of success for describing the scenes, it is still not interactive and does not exploit the user's feedbacks. Moreover, WordsEye relies on its huge offline rule-base and data repositories containing different geometric shape which lacks an automatic annotation task.

F. *CONFUCIUS*

This is a multi-modal text-to-animation conversion system that can generate animation from a single input sentence containing an action verb and synchronize it with speech [16]. It is composed of several modules with different tasks to accomplish such as Knowledgebase, Language processor, Media allocator, Animation engine, Text-to-speech engine, Story narrator, Synchronizer. As seen in Figure 6, CONFUCIUS supports lip synchronization, facial expressions, and parallel animation of the upper and the lower body of human models.



Figure 6. Output animation of "John put a cup on the table."

Generally, CONFUCIUS is not interactive in a sense that it does not let the user modify the generated animation. In addition, only a single sentences in each input is allowed and only a restricted format of the input sentences (i.e., one action verb per sentence and only simple verbs are considered) is permitted, hence the user is restricted in expressing the intended description. Table 1 gives an overall comparison of surveyed text-picture systems focusing on their common features.

Table 1. Comparison of text-to-picture systems

System	Syntax Analysis (NLP)	Semantic Analysis (NLU)	Input Modality	Output Modality
Story picturing engine	Bag-of words	Semantic association	Paragrap h	2D Picture
Text-to-picture synthesizer	POS tagging	Association Semantic Role labeling	Typed Text	2D Picture Collage
Word2Image	Bag-of words	Semantic association	Words being pictured	2D Picture Collage
Chat with Illustration	Bag-of words	Semantic association	Typed Short text	2D Picture
WordsEye	Statistical parsing	Dependenc y Parsing	Story	3D Pictures
CONFUCIUS	Dependen cy parsing	Lexical Visual Semantic	1 Sentence	3D Animati on, Audio

IV. SYSTEM ARCHITECTURE

We propose a new system to improve the learning process of the Arabic language for children with LD and for non-native Arabic learners. Our system aim is to generate personalized multimedia contents dynamically through providing an automatic mechanism to query for a desired topic based on semantic content analysis. The system will statistically learn the preferred customization per student and will generate automatically customized tutorials. The dynamic tutorials would become more time efficient due to the machine learning process of the automated system. Figure 7 shows a high level architecture of the proposed system. It consists of generating multimedia-based tutorials by mining Arabic text, and it uses a core multi-domain ontology that exploits existing Arabic natural language processing technologies including morphological analysis, logical rules and discretization.

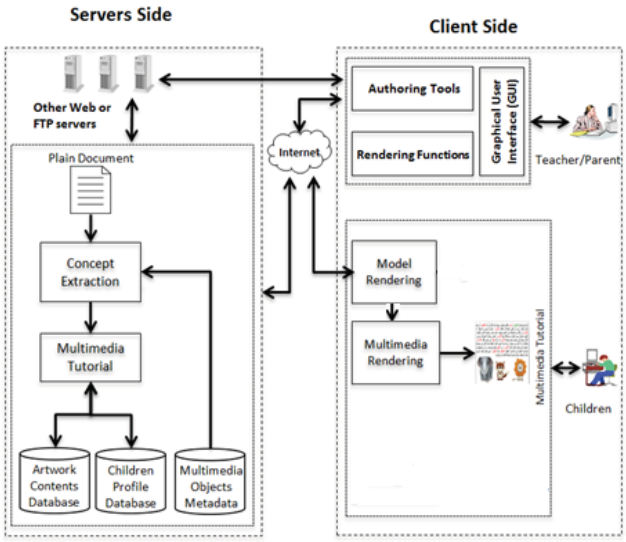


Figure 7. Proposed System Architecture

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The first input is the educational text that will be processed using segmentation, stemming, and analyzing. We parse text with the Stanford Parser that aids to retrieve sentences containing pairs of keyword entities and represents them using a dependency graph. The objective is to identify the actors (i.e., lion, elephant, tiger), the objects (i.e., water, plant), and the actions (i.e., attack, live, eat) in the story text. Each retrieved sentence is analyzed to extract the raw relationships between keywords entities. The extracted relationships and entities will be semantically mapped and validated according to the domain description captured in an ontology model. All raw relationship instances are sent as semantic queries in order to retrieve the available multimedia instances in our ontology (i.e., the knowledgebase). Complimentary queries will sent also to the Google image database through Google APIs to fetch the corresponding multimedia elements. Hence, the teachers can then select the appropriate multimedia elements that are suitable to customize the tutorial based on the needs of each child in the classroom.

A. Proposed Approach

In this section, we describe thoroughly the approach for mapping Arabic simple stories to images. This approach is split into two main parts, represented in Figure 1: (1) Story text processing and image retrieval using keyword-based search, containing the steps 1, 2 and 3. (2) Image ranking using automatic captioning process and sentence similarity, containing the steps 4, 5, and 6.

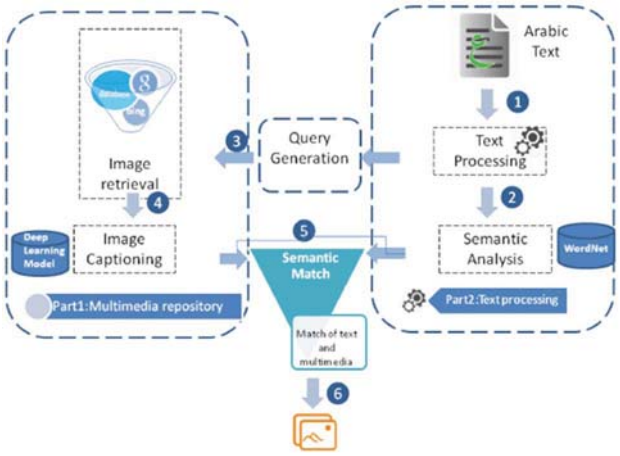


Figure 8. Proposed Approach for text-image mapping

To achieve this functionality, we performed the following tasks:

1. Collecting Arabic simple stories in the domain of animals.
2. Processing Arabic text using different NLP tools for Arabic.
3. Generating SQL queries and search engine queries to retrieve images and creating a database to store the mapping of keywords to images to serve as preliminary image pool.
4. Captioning retrieved images using a Convolutional Neural Network (CNN) pre-trained model.
5. Matching semantic aspects by retrieving a set of images with initial input text.
6. Validating and ranking of the retrieved images using sentence similarity.

To perform those tasks, we used several open source tools such as MT tool [17], CNN/LSTM image captioning model [18], etc. In the following, we present in detail the different steps.

B. Description of the Approach Steps

1) Keyword Extraction:

First, story text is split into sentences and some preprocessing steps are made such as segmentation, stop-word removal, and part-of-speech tagging, etc. Second, we select relevant single word tokens as keywords for each sentence and we translate them to English using this MT translation tool [17], [19]. We organize the keywords in the order that they appear in the text to preserve their role within the text.

2) Query Formulation:

In this step, we formulate keyword-based queries to retrieve candidate images for them. A useful heuristic for finding better representative images for the characters and the events in search engines is to concatenate the extracted keywords including verbs as a single query for each sentence. A standard method of multiple queries is also employed. The combination of single query and multiple queries is also employed.

3) Image Selection:

The retrieved images using the concatenation of keywords are downloaded, saved locally and thus prepared for further pre-filtering and captioning process in the next step.

4) Image Captioning

The prepared set of images is ready for going through this step in order to be captioned by a deep learning model. Image captioning via deep convolutional neural network (CNN), recurrent neural network (RNN) and long short-term memory (LSTM) have witnessed significant improvements in recent years [20]. Deep CNNs can fully represent an image by embedding it into a fixed-length vector. Then, RNN, especially LSTM [21], decodes the fixed-length vector to a desired output sentence by iterating a recurrence relation [22]. We used a pre-trained model as a fine-tuned checkpoint which has been trained over 3 million iterations using the MSCOCO dataset [23].

5) Sentence Similarity

The sentence similarity is requested in order to make the matching between initial keywords and captions. We apply sentence similarity after the Machine Translation (MT) process to guarantee basic textual information for the matching process. The obtained English captions are then MT processed into Arabic. The resulting Arabic captions are compared with keywords to find out which images are kept for final representation. We use sentence similarity to estimate values for similarity. At this early stage, is not clear yet which similarity function best fit within our case. It is a hard task and most of the metrics fail in identifying the similarity between all variations of text as argued in [24] and [25]. For sentences similarity task, we adopt a standard approach to compare the similarity between sentence pairs by computing a cosine similarity [26] between two sentences. Besides, we employed semantic similarity using WordNet published in these open resources [27], [28] for English sentences. Only images with similarity greater than a user-defined threshold are considered for further selection.

6) Image Ranking

Based on obtained similarity values, we select sentences that have higher similarity values with the keywords. Associated top-10 images are ranked and eventually shown to the user/teacher, whenever he clicks the option for that.

7) Image Evaluation

This step is based on manual evaluation for each relevant image where the user can give a rank from 1 to 5. The system uses this value as an initial score for the final score calculation.

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C. System Operation

We briefly describe the system work flow. As depicted in Figure 9, our proposed system works as follows: User or teacher inputs a story text in Arabic using simple sentence structure, here "يركض النمر وراء الغزلان". Then, key words are extracted using main text processing steps; text segmentation, pos-tagging, and stop-words removal. After that, the MT process to English is applied. Then, the user clicks on "Multimedia search"' either local search or Google search. After that, retrieved images are displayed. The user can validate them to be stored in a local database, in case they are not yet validated. Finally, the user can display top-10 ranked images and use them to explain the main ideas and objective of the given each sentence in a story.

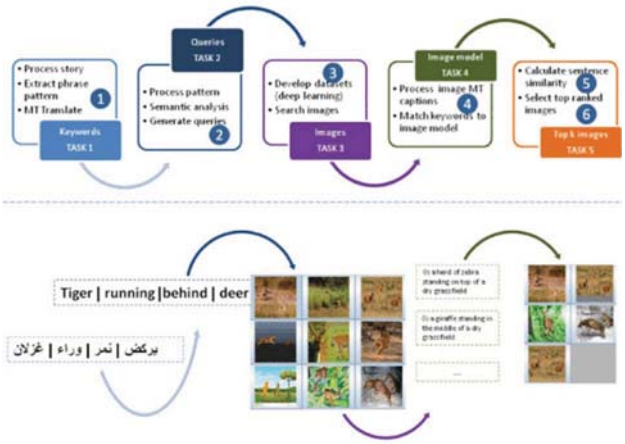


Figure 9. Systematic flowchart of the proposed system

V. EXPERIMENTS AND EVALUATIONS

This section discusses the evaluation of main components of the proposed system. These components are (i) keyword based image search; (ii) sentence similarities; and (iii) the image retrieval and ranking. The goal of the evaluation is to test whether the images generated by our proposed system are able to accurately convey the main characters of the stories. We distinguish here four different cases: AW, AWC, EW, and EWC as indicated in Table 2.

Table 2. Description of four cases used in evaluations

Cases	Descriptions
AW	Arabic Words without Captions
AWC	Arabic Words with Captions
EW	English Words without Captions
EWC	English Words with Captions

A. Experimental Setup

In the experimental setup, we start by following settings:

- Input text data set, we input 30 short simple stories in the domain of animals to our system. In the future, it is planned to set up a corpus for Arabic children's stories.
- Database has been set to store images and keywords and their mappings. For each keyword, we store 10 images and set a manual ranking from 1 to 5. Once we rank an image, we validate it to be shown to the learner. This step is necessary to preserve a safe learning environment for the learner. Image captions have been generated using a CNN pre-trained model [18]. Obtained captions are available in English only and been MT translated in Arabic. These data are also stored in the database.
- Search engine, we used Google image search [29] to fetch additional images. It is worth mentioning that we use Google image search in our prototype for illustrative purpose only. In the future, it is planned to set up a dataset of appropriate educational images for better learning and understanding.
- Deep learning model, we use im2txt, which is a pre-trained model for tensorflow published on github [18]. It is a model developed by Google that takes an image as input and creates a caption for it. Basically, a deep CNN is used to encode images to a fixed-length vector representation. It is followed by a LSTM network which in turn takes the encoding and produces a caption.
- MT translation: Extracted keywords are MT processed to English using the QCRI MT-tool [18]. Furthermore, captions are MT processed to Arabic using the same tool.

B. Evaluation Criteria

We define following evaluation metrics based on general notion of positive and negative human judgments in image retrieval:

- Relevance a metric of retrieved images that most represent a specific keyword/story.
- Precision [30] is a proportion of relevant retrieved images to all retrieved images.
- Top10 are relevant images having 10 highest scores.

The similarity score produced by these measures has a normalized real-number standing within the unit interval. We have performed evaluation experiments using these metrics on a set of 30 stories to analyze user satisfaction with the output and the impact of the automatic captioning process on similarity measures and consequently on the ranking of top-10 images.

C. Evaluation of Image Relevance

The relevance and the suitability of each image were obtained through expert user. User was provided with 30 stories together with their output images from our proposed system. He was asked to judge if the images were a representation of the main subject of the topic and provide a rating on a scale of 0 (completely unsuitable) to 1 (very suitable). The obtained results show that the average precision of our system is 44% for the AW-version compared to 49% for the EW-version. The user satisfaction with top10 is 38% for the AW-version whereas it is equal to 52% for the EW-version. We adopt the precision as the evaluation criterion by comparing the output each time a new story is entered.

D. Evaluation of Image Captioning

We applied an automatic captioning process for top-10 retrieved images which are treated as a preliminary set of images. The obtained results show that the average precision value for top-10 under the AWC-version is 46% compared to 56% for the EWC-version. The captioning process is done automatically and saved us time and resources, however, the results are weak; the model creates meaningful captions for images with common objects. But, it fails when more abstract objects or similar objects in shape are present. This could be improved by considering different techniques for training.

E. Evaluation of Translation Quality

For evaluating the quality of MT translation of image captions from English to Arabic, each sentence pair is judged by two humans whether the sentence pair is semantically coherent or not. The users gave a similarity score between 5 and 0, where 5 means a good translation and 0 means that there is no semantic

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similarity between the pair of sentences. The obtained results showed overall 15 % of the total sentence pairs are judged to be positive examples means they are semantically equivalent. Consequently, the value indicates relatively poor performance in judging similar pairs. Thus, it adversely affects the overall accuracy. For the quality of translation, we remark that translations of image captions from English to Arabic have led to low sentence semantic similarity values with the input stories. These results are not encouraging to use MT translation tool for our future work.

F. Evaluation of Text-Multimedia Mapping

We target at enhancing our current proposed system that is still under development and improvement. We are currently processing sentences with simple structure where stop words are eliminated. Then, keywords are extracted and searched first in the local database and their corresponding images are presented. Using the concatenation of all keywords produced often more relevant for delivering the meaning of the whole story and consequently has led to better performance than separate keywords. Top10 images are also an option to save time in looking over all relevant images. Whenever the retrieved images are not relevant, the user can address a query to Google Search. A validation step for new images is required in this case to keep a safe learning environment.

VI. CONCLUSION

We proposed an adaptive multimedia learning system for assisting children with Arabic LD to enhance their understanding skills of Arabic vocabulary and sentences. In this paper, we introduced an approach to automatically convert simple Arabic children's stories to the best representative images from Google image search using deep learning model for image captioning. The system can be effectively used by instructors to provide children with LD the opportunity to learn with contents designed with multimedia which would maximize the learning process and reduce the administrative and teaching overheads. The system can be used in schools' settings as well as at home where parents can assist their children to review the materials they studied in schools.

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